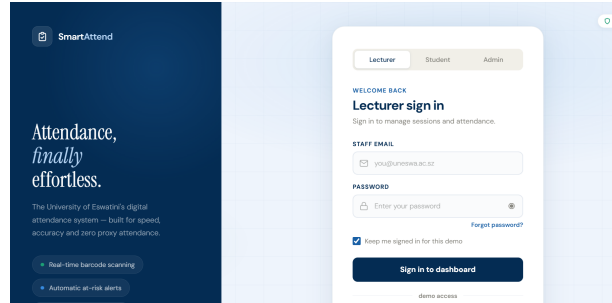


SmartAttend

Smart Attendance System

Barcode scanning for
lectures and tests.

University of Eswatini - CSC392



Project overview and team

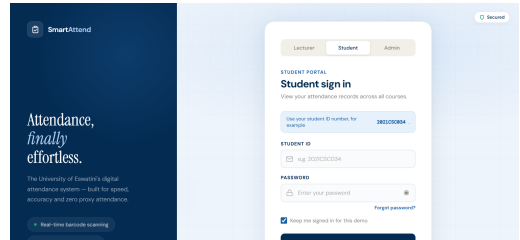
PROJECT INTRODUCTION

Brief description

SmartAttend is a software-hardware integrated system that records attendance using student ID barcode scanning. It supports registered, unregistered, and unknown attendance cases.

Team responsibilities

Wakhile Mabuza	Team Manager
Sean Rodrigues	System Architect
Njabulo Simelane	UI/UX Designer
Sakhiwo Magongo	Programmer



Multi-role sign-in: lecturer, student, and admin.

Problem statement

WHY THE PROJECT IS NEEDED

Current problems

- ▶ Manual attendance sheets are time-consuming.
- ▶ Students can sign for absent friends.
- ▶ Attendance records can be lost or damaged.
- ▶ Reports are difficult to generate quickly.
- ▶ Test attendance is hard to verify accurately.

Impact

- ▶ Inaccurate participation records.
- ▶ Administrative inefficiency.
- ▶ Weak monitoring of student attendance.
- ▶ Poor evidence for at-risk student follow-up.

75%

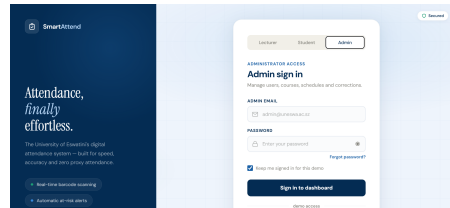
Attendance threshold can be monitored automatically.

Aims and objectives

WHAT SMARTATTEND MUST ACHIEVE

Main aims

- ▶ Automate attendance recording.
- ▶ Reduce proxy attendance.
- ▶ Provide real-time attendance tracking.
- ▶ Integrate barcode scanning with the software system.
- ▶ Produce reports for lecturers and administrators.



Role-based access protects system functions.

Proposed solution

CORE SYSTEM CAPABILITIES

SmartAttend will

- ▶ Scan student ID barcodes using a scanner or webcam.
- ▶ Verify student information from the database.
- ▶ Check whether the student is enrolled in the course.
- ▶ Record lecture and test attendance.
- ▶ Flag unregistered or unknown attendees.
- ▶ Generate attendance reports and CSV exports.

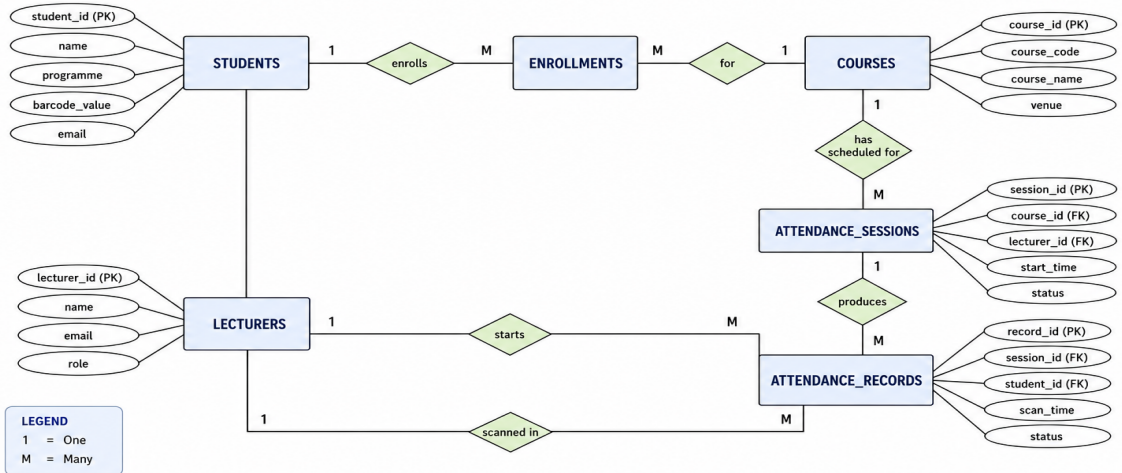
Primary users

- ▶ **Lecturer:** opens sessions, scans students, views reports.
- ▶ **Student:** views attendance and barcode.
- ▶ **Admin:** manages students, courses, schedules, and corrections.

ER diagram

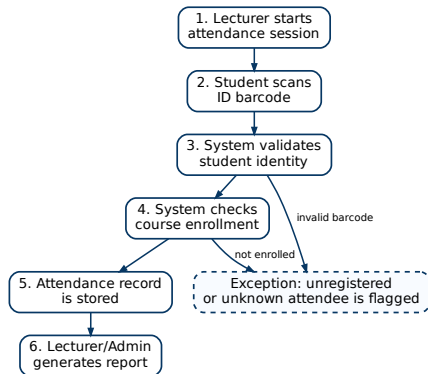
DATABASE DESIGN

The ER diagram below shows the main entities in SmartAttend and how they relate to each other.



System flow

HOW ATTENDANCE IS CAPTURED



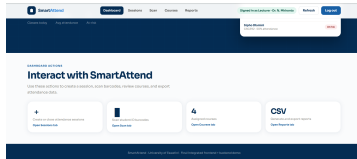
Step-by-step process

- ▶ Lecturer opens a lecture or test session.
- ▶ Student scans an ID barcode.
- ▶ System validates identity and enrollment.
- ▶ Attendance is stored in the database.
- ▶ Invalid or unenrolled scans are flagged.
- ▶ Reports are generated when needed.

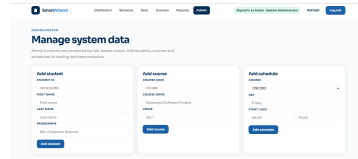
Lecturer and admin interface screenshots

IMPLEMENTED SCREENS

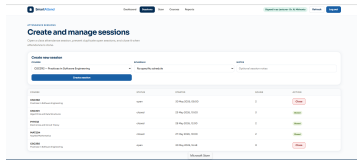
Lecturer Dashboard



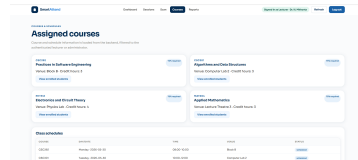
Admin Panel



Session Management



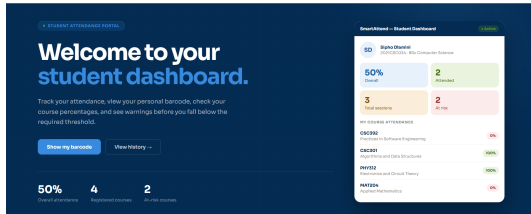
Courses and Schedules



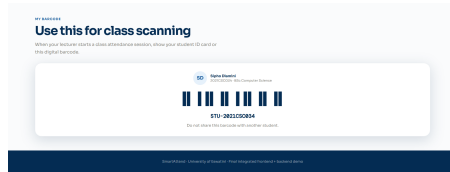
The interface separates lecturer actions from protected administrator data management.

Student portal and reporting screenshots

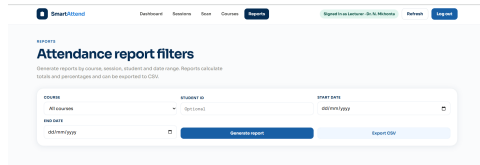
STUDENT AND REPORTING FEATURES



Student dashboard shows overall attendance, course percentages, and at-risk courses.



Digital barcode for class scanning.



Report filters by course, student, and date range.

Technologies, progress, and conclusion

FINAL SUMMARY

Technologies	Progress	Expected benefits
▶ React frontend ▶ Node.js and Express backend ▶ PostgreSQL database ▶ GitHub collaboration ▶ Barcode scanner or webcam	▶ Project approved ▶ SRS completed ▶ ER diagram completed ▶ Database design completed ▶ Implementation started	▶ Faster attendance capture ▶ Reduced proxy attendance ▶ Improved data accuracy ▶ Automated reporting ▶ Better lecture and test monitoring

SmartAttend makes attendance faster, safer, and easier to manage.